CHAPTER II.

THE BRAIN AND NERVOUS SYSTEM.

Under the microscope the diameter of blood-disks is one four-thousandth part of an inch!

Hundreds of millions of air-sacs in the lungs, miles of tubing in the blood-vessels and capillaries, and millions of cells occupied in forming the brain; phrases, these, which, however true, bring no sense of realization to the mind seeking to learn the grand machinery of life, except to those familiar with the dissecting-room, or conversant with works devoted exclusively to the delineation of anatomy and physiology.

Therefore, in this volume, all minutiæ of description which applies to the original formation of tissues is withheld, and the simple statement is given that all the different portions of the body, whether bone, muscle, membrane, cartilage, or nervous matter, are constructed from minute cells, the primary form of which is a sac of membrane of infinitesimal attenuation and nuclei of microscopic size.

Millions of these, flattened, elongated, rounded, or pressed into spheroids, go to make up a single fibre of muscular or nervous matter. Bundles of fibres, countless in multitude, are bound together to form a mass of tissue; whilst the human body includes the totality of all tissues, organs, systems, and apparatuses.